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Electromagnetic waves and their effects on the human organism

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Abstract: By studying electromagnetic waves and their effects on the human body, it was found that modern people, especially those living in big cities, are more affected by electromagnetic waves. Electromagnetic waves cause unwanted changes in the human body: the central nervous system is overloaded, the response to stimuli is delayed, memory deteriorates, and at the same time, it causes various degrees of depression, high excitement, additional irritability, sleep disturbance, mood deterioration, dizziness, weakness, etc. causes situations.

Keywords: Electromagnetic waves, High-voltage lines, Electric field, Magnetic field, Blood-vascular system, Duration of exposure to radiation.

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1. Introduction

Among the countries of the world, important work is being done in the direction of reducing the emission of greenhouse gases and minimizing the external effects on the living organism. In recent decades, an increase in the average temperature in the environment has been observed. At the same time, due to the development of science and technology, the increase of new electronic devices has also increased the radiation of electromagnetic waves [1]. The damage caused by a large number of enterprises surrounding us to the environment and human health is estimated by special mathematical models [2]. From those studies, it is known that the pressure of the environment on the human body is increasing day by day. Although electromagnetic waves (frequency above 300 Megahertz) have a high impact on the human body, people always have to use them to create a comfortable lifestyle. For health, it is necessary to follow some rules to reduce the effect of these waves or to protect them.

Electromagnetic waves consist of electric and magnetic fields propagating at a high ($3 \cdot 10^8$ m/s) speed [3].

2. Experimental detail

Power transmission lines are capable of radiating those powerful electromagnetic waves. High-voltage transmission lines are found everywhere, posing real threats to the life and health of people living in close proximity. Therefore, it is undesirable to live near high voltage transmission lines.

- Emitters of electromagnetic waves can be divided into the following groups: voltage lines,
- high voltage transformer transmitters,
- radio and TV tower transmitters, power stations,
- household appliances,
- mobile devices.

The highest radiation sources of electromagnetic waves are observed under high-voltage lines, near high-voltage transformers and high-voltage distributors.

Scientists hypothesize that electromagnetic waves created by humans can affect both plant life and living things. For example, the process of protein synthesis in plants and even the structure of DNA can be disturbed, and as a result, the plant's development can be weakened. However, it is still not known how the degree of influence of those effects depends on the frequency, intensity and duration of the effect. The main issue is whether electromagnetic waves harm human health. On the other hand, scientists come to the conclusion that electromagnetic waves affect the human body and living organisms in general.

There are certain biocurrents in the human body. Those microcurrents can be generated as a result of various biological processes: for example, from the digestion of food to the moment of active thinking in the brain. It is assumed that the low-frequency electromagnetic field creates additional currents in the human body, which in turn create additional tensions in the muscles and nervous system. A low-frequency electric field can affect the flow direction of natural microcurrents in a living organism. For example, it can reduce the speed of transmission of intercellular signals in the nervous system or disrupt their order.

1. One of the sources of strong radiation in big cities is the subway. In addition, trolleybuses, trams, elevators of high-rise buildings also radiate electromagnetic waves. When the value of the magnetic field is greater than 8 Tl, the effect of external electromagnetic waves on the ions in the electrolyte in the blood is strengthened, and as a result, the flow of blood in the vessels is weakened. Therefore, the rhythmic beating of the heart is disturbed - the duration of the pauses increases, the risk of arrhythmia increases, and even the fibrillation of the ventricles in the heart changes. The increase in the power of various artificial sources: radio communication, radar, high-voltage electrical transmitters, microwave ovens, computers, notebooks, televisions, transport, etc., leads to an increase in the level of artificial electromagnetic waves in cities and, as a result, creates an undesirable effect on human health. In addition to the above, electromagnetic waves cause the following damages in the blood-vascular system of the human body:

1. Changes occur in the blood-vascular system: heart rate decreases and arterial blood pressure changes.

2. The morphological composition of the blood is disturbed: the number of leukocytes, reticulocytes, acidophilic granulocytes decreases.

All organs in the human body have their own vibrational frequency. Due to this, a personal electromagnetic field is created around each of us. When this area is exposed to the effects of external electromagnetic waves, a person feels physically ill. Long-term exposure to strong electromagnetic waves can also cause illness over time. The human nervous system is very sensitive to electromagnetic waves. Due to its influence, a person may experience fatigue and nervousness, sleep problems, and memory may deteriorate. Even severe depression can occur. Usually, a young and healthy body easily copes with small doses of electromagnetic waves. However, long-term exposure to electromagnetic radiation is considered dangerous for pregnant women, people with chronic diseases or the elderly. Strong electromagnetic waves have also been observed to be harmful to plants. Where powerful telecommunications transmitters and antennas are installed, entire vegetation can dry out and die. In order to protect yourself from the deadly effects of electromagnetic sources, you should know what electromagnetic devices are covered and pay attention to their placement in the home environment:

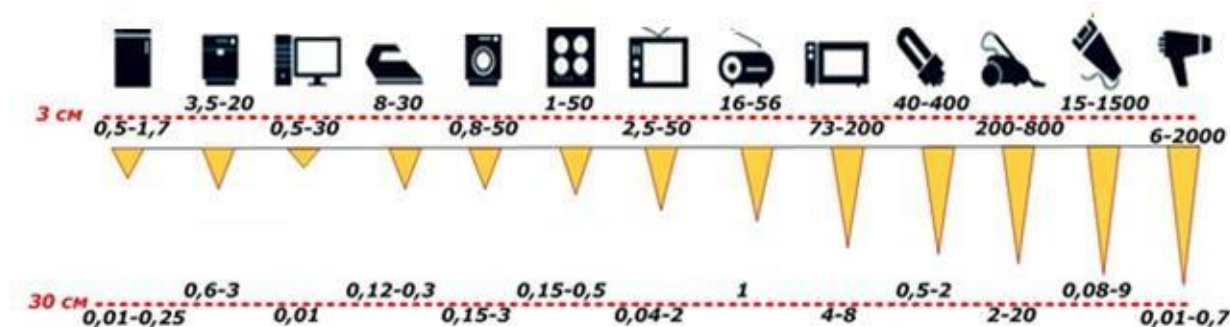
- Electrical appliances cannot be placed in the same part of the house. They should be placed in such a way that the electromagnetic fields they create do not reinforce each other;
- Electrical devices, especially telephones, should not be placed near the dining table, work table, rest and sleeping place;
- Children's rooms should be isolated from electromagnetic devices as much as possible. So, regular operation of radio and electrically controlled toys, tablets, smartphones and notebooks cannot be allowed in those rooms;
- Electrical devices should be turned off one by one, not at the same time. Therefore, the mobile

phone, computer, ultra-high frequency (microwave) oven, TV should be used at different times;
- A radiotelephone source creates a constant magnetic field at a distance of 10 m around itself. It is recommended to remove this device from the bedroom and the desk.

When using household devices such as microwave oven, air grill, refrigerators with "no frost" system, kitchen air extractor, electric heaters, TVs, it is necessary to take into account that they have a high electromagnetic field.

The higher the frequency of electromagnetic oscillations, the higher the danger to human health. At a distance of half a meter from the remote-controlled household appliance, the electric field strength should not exceed 500 V/m at a frequency of 50 Hz. For 25 V/m, the allowed frequency is 0.3-300 kHz, and in the high-frequency range 30-300 MHz, only 3 V/m is allowed. As the intensity of radiation increases, so does its danger. A base station antenna with a power of 20 W has a greater impact on human health than a mobile phone with a power of 1 W if placed near it. If the antenna of the base station is located at a considerable distance from the operator, but the power of its transmitter at the receiving point is, for example, 1 mW, then it is already dangerous to use a mobile phone.

As the frequency range expands, so does the radiation hazard. The vast majority of household appliances have a very narrow frequency emission range. For such equipment, the "SanPiN 2.1.2.1002-00" system is applied. Equipment with a more complex design and a wide frequency range within a few gigahertz are separately regulated by the "SanPiN - 2.2.2/2.4.1340-03" system. The radiation of the remote controls of these devices is also strong: about 20 times the norm for electric radiation, and 40 times for magnetic radiation.



An important factor in the impact of electromagnetic waves on the human body is the duration of exposure to radiation. Therefore, if a person likes to talk on a mobile phone for a long time, then it is better to use a headset so that the phone is not next to his head all this time. Otherwise, the brain may be exposed to electromagnetic radiation for a long time [4, 5].

Russian industry also produces measuring instruments for domestic needs. For example, it is possible to determine the parameters of electromagnetic radiation from household appliances with a "pulse meter". The working frequency range of that device is between 20-2000 Hz. With its help, you can easily determine the security level of home computers, microwave ovens, smartphones and other household appliances.

If there is a high-voltage power line near the residential building or if there are underground electric cables running nearby, then it is necessary to take into account that there is a danger to health. If it is not possible to call a special mobile laboratory to measure the doses of radiation sources, then try to make the measurements yourself. In this case, the device "RADEKS EMI 50" operating in the range of 47-53 Hz can be used.

If we do not use common household electrical appliances properly in our daily life, those appliances can become a source of health hazards. For example, although modern smartphones are safe, when they are connected to a base station, the level of electromagnetic radiation increases significantly within minutes. This happens especially right after the number is dialed. Therefore, you should not hold your phone close to your head during that time.

Usually, the radiation emitted from the computer is within normal limits, but doctors recommend

taking a short break every 45-60 minutes. It is not recommended to hold notebooks in your arms, because in this case the waves of the computer have a stronger effect on the human body. You may be exposed to electromagnetic fields when using hair dryers. When using that device, keep them away from your head. Electric kettles, juicers, coffee makers, washing machines, air conditioners and all other appliances with electric motors also emit electromagnetic waves. Various telecommunications equipment such as Wi-Fi are more likely to be sources of danger. Therefore, it is recommended to turn off modems at night.

In apartments, near electric meters and distribution boards, the permissible level of the electromagnetic field should not be exceeded. To reduce the impact of electromagnetic waves in houses and apartments, we must follow simple rules:

- place sleeping and working places as far as possible from radiation sources,
- should not use broadband frequency range equipment;
- when using sources of electromagnetic radiation, it is advisable to take them intermittently,
- use less high-frequency equipment,
- monitor the technical condition of household appliances and follow the manufacturers' recommendations;
- periodically measure the doses of electromagnetic waves in residential buildings.

It should be remembered that, as a rule, a person does not feel the negative effects of electromagnetic waves. However, the consequences of those effects become apparent after a certain period of time and it is already too late. Therefore, every user should take care of health so that the situation does not become critical!

3. Conclusion.

From the research conducted on "Electromagnetic waves and their effects on the human body", it became clear that electromagnetic waves really affect all living organisms and also plant life. It slows down their development process and the process of protein formation. The vibration frequency of all organs in the human body changes, the normal functioning of the blood-vascular and nervous system is disturbed, the number of chronic diseases increases, headaches occur, fatigue is observed, and in general, the general mood deteriorates. Therefore, it is necessary to use the electronic devices, communication tools, household appliances and high-voltage power lines that we use in our daily life correctly.

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A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top.

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A graphic of a water splash or wave, rendered in a soft, painterly style, located on the right side of the lower half of the cover.